Introduction to MySQL

MySQL

It is freely available open source Relational Database Management System (RDBMS).

MySQL was created and is supported by **MySQL, AB,** a company based in Sweden (www.mysql.com). The chief inventor of MySQL was Michael Widenius. MySQL has been named after his daughter **My**.

Features of MySQL

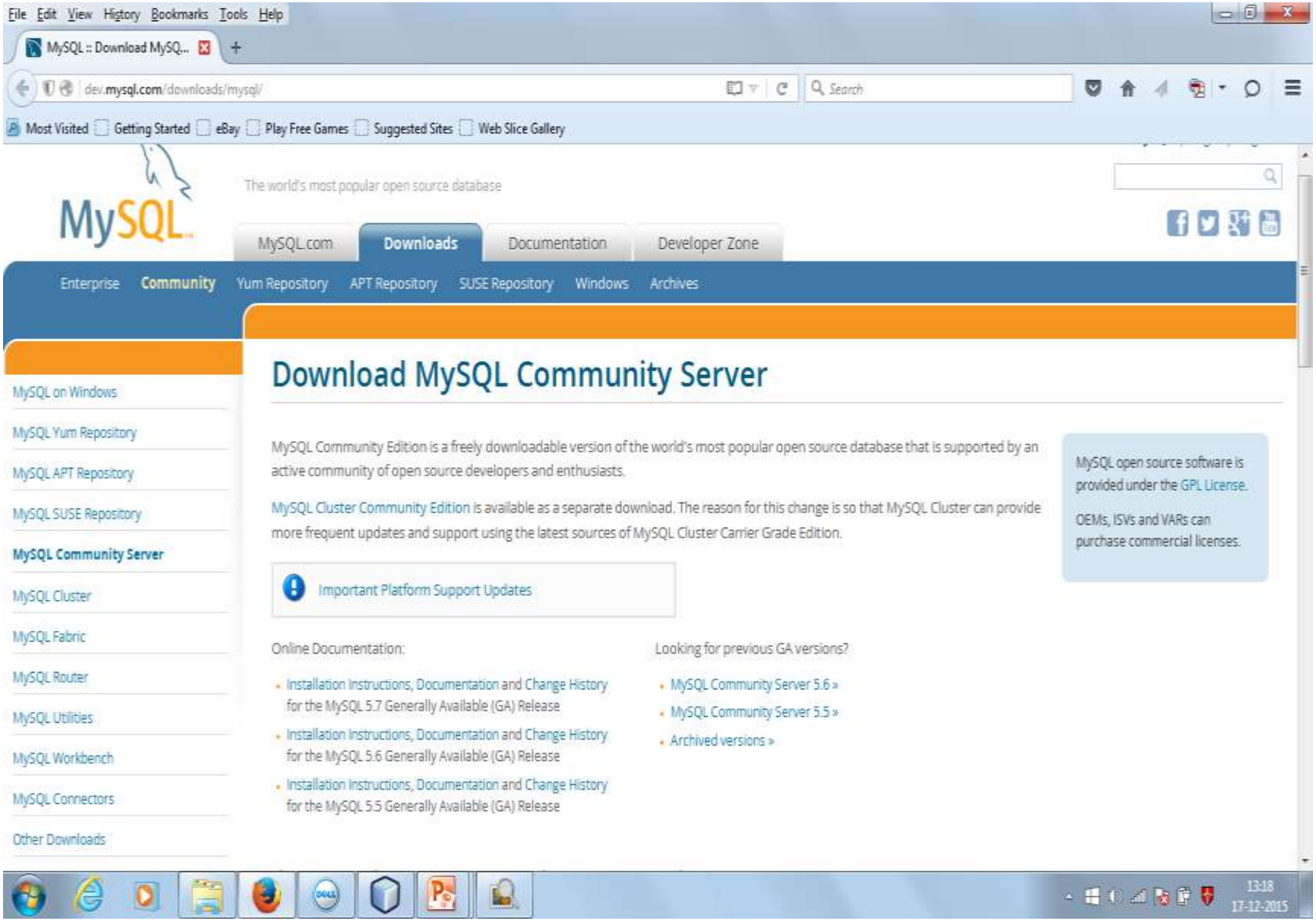
- MySQL is released under an open-source license so it is customizable. It requires no cost or payment for its usage.
- MySQL has superior speed, is easy to use and is reliable.
- MySQL uses a standard form of the well-known ANSI-SQL standards.
- MySQL is a platform independent application which works on many operating systems like Windows, UNIX, LINUX etc. and has compatibility with many languages including JAVA , C++, PHP etc.
- MySQL is an easy to install RDBMS.

DOWNLOADING MySQL

Installation file for MySQL may be downloaded from the link:

http://dev.mysql.com/downloads/mysql/

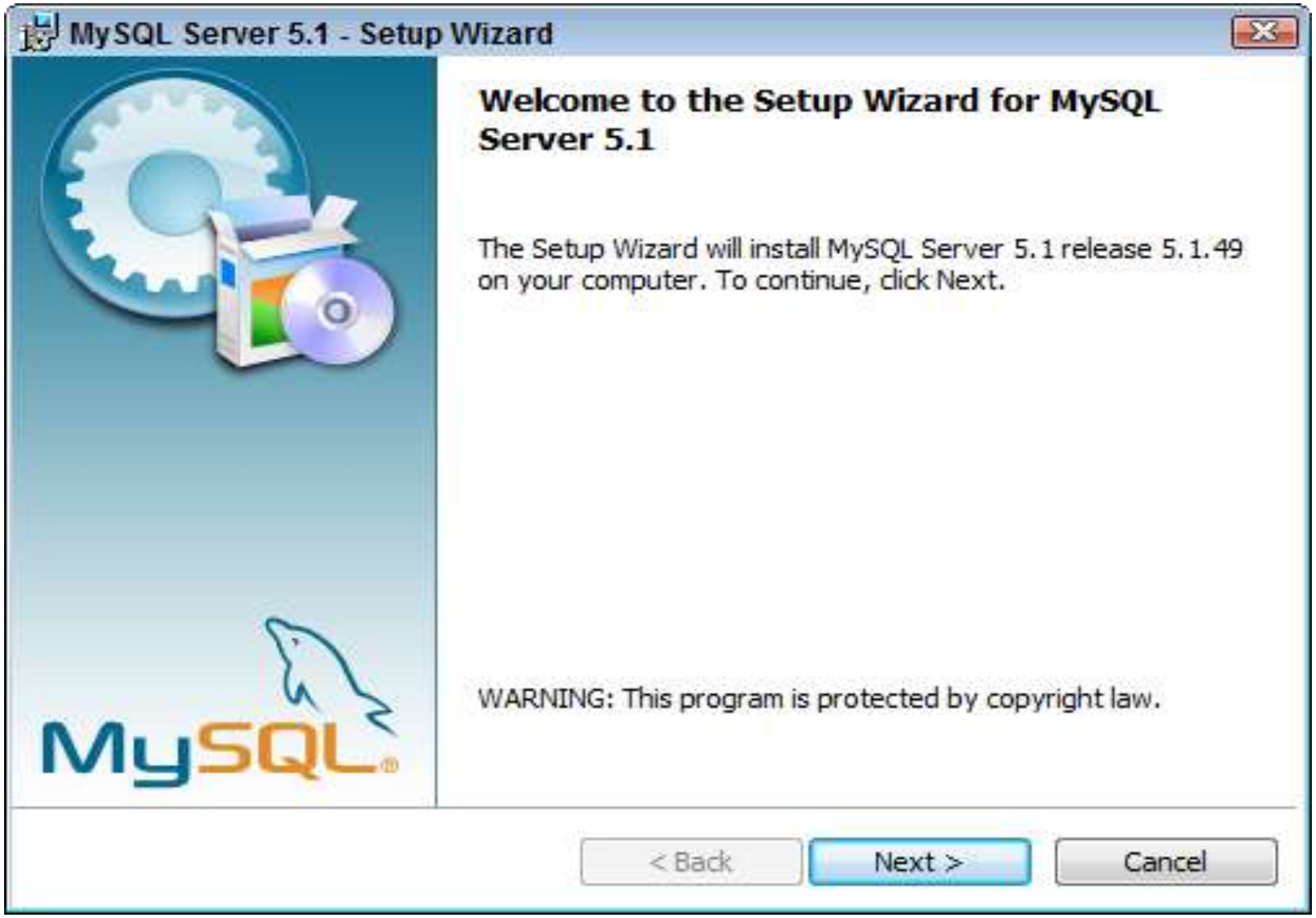
(Choose appropriate download link as per the operating system)

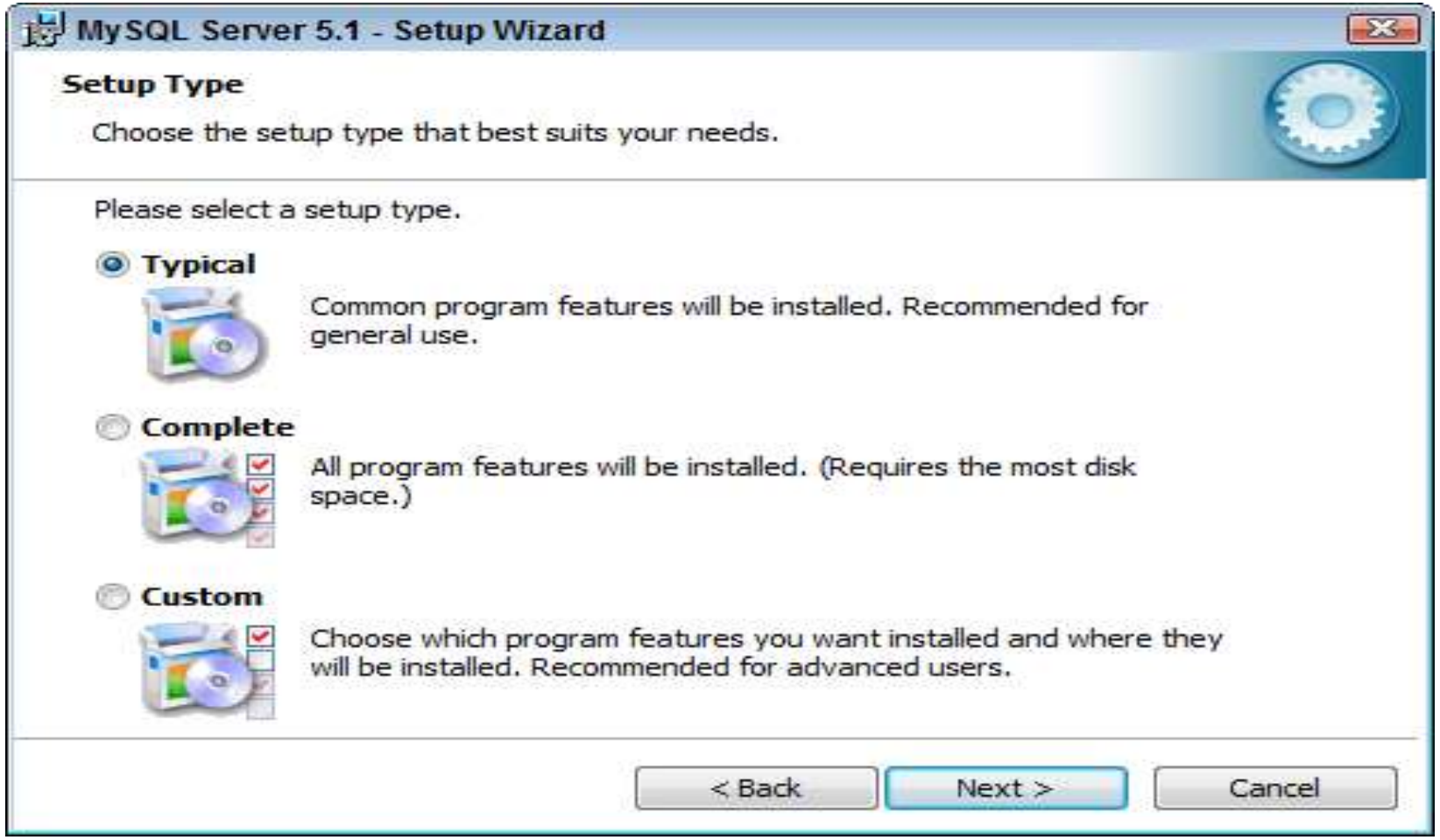

File Edit View History Bookmarks Tools Help
MySQL :: Download MySQ...
dev.mysql.com/downloads/mysql/
Search
Most Visited
Getting Started
eBay
Play Free Games
Suggested Sites
Web Slice Gallery
MySQL
The world's most popular open source database
MySQL.com
Downloads
Documentation
Developer Zone
Enterprise
Community
Yum Repository
APT Repository
SUSE Repository
Windows
Archives
MySQL on Windows
MySQL Yum Repository
MySQL APT Repository
MySQL SUSE Repository
MySQL Community Server
MySQL Cluster
MySQL Fabric
MySQL Router
MySQL Utilities
MySQL Workbench
MySQL Connectors
Other Downloads
Download MySQL Community Server
MySQL Community Edition is a freely downloadable version of the world's most popular open source database that is supported by an active community of open source developers and enthusiasts.
MySQL Cluster Community Edition is available as a separate download. The reason for this change is so that MySQL Cluster can provide more frequent updates and support using the latest sources of MySQL Cluster Carrier Grade Edition.
Important Platform Support Updates
MySQL open source software is provided under the GPL License.
OEMs, ISVs and VARs can purchase commercial licenses.
Online Documentation:
Installation Instructions, Documentation and Change History for the MySQL 5.7 Generally Available (GA) Release
Installation Instructions, Documentation and Change History for the MySQL 5.6 Generally Available (GA) Release
Installation Instructions, Documentation and Change History for the MySQL 5.5 Generally Available (GA) Release
Looking for previous GA versions?
MySQL Community Server 5.6 »
MySQL Community Server 5.5 »
Archived versions »
13:18
17-12-2015

INSTALLING MySQL

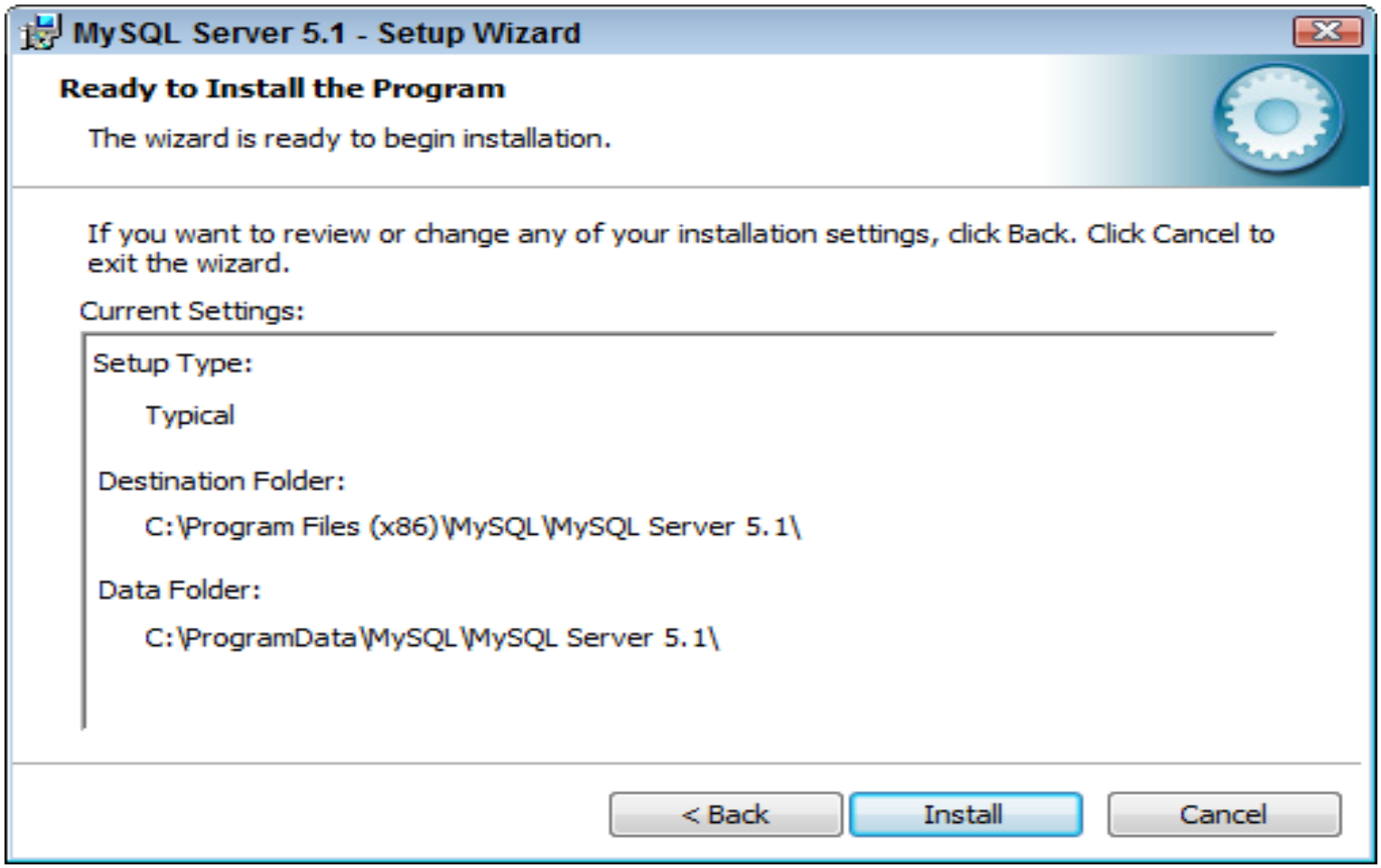
INSTALLING MySQL

After the installation file has finished downloading, double-click it, which begins the MySQL Setup Wizard.

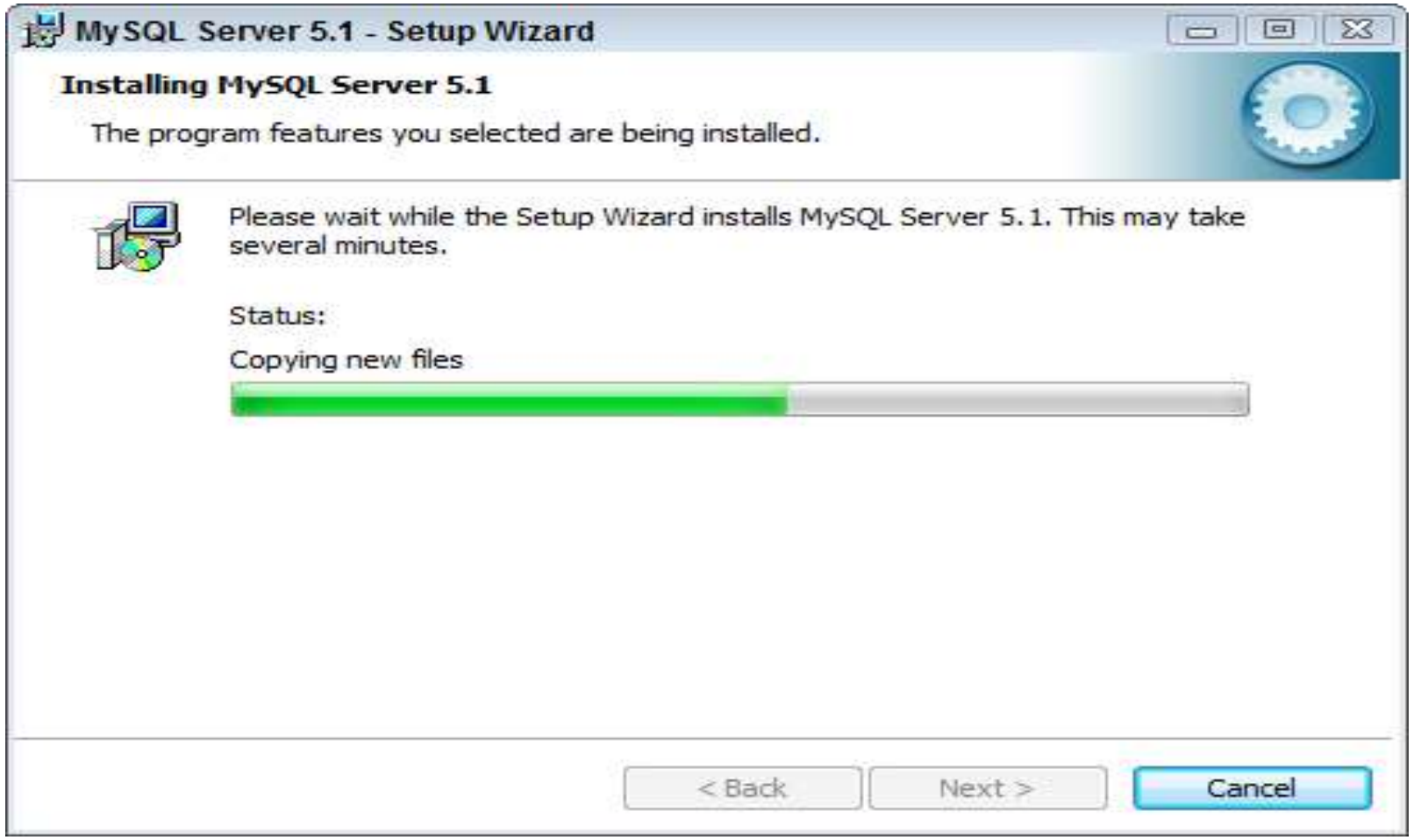

MySQL Server 5.1 - Setup Wizard
Welcome to the Setup Wizard for MySQL Server 5.1
The Setup Wizard will install MySQL Server 5.1 release 5.1.49 on your computer. To continue, click Next.
WARNING: This program is protected by copyright law.
MySQL
< Back
Next >
Cancel



Select the Typical installation option
Click [Next] button



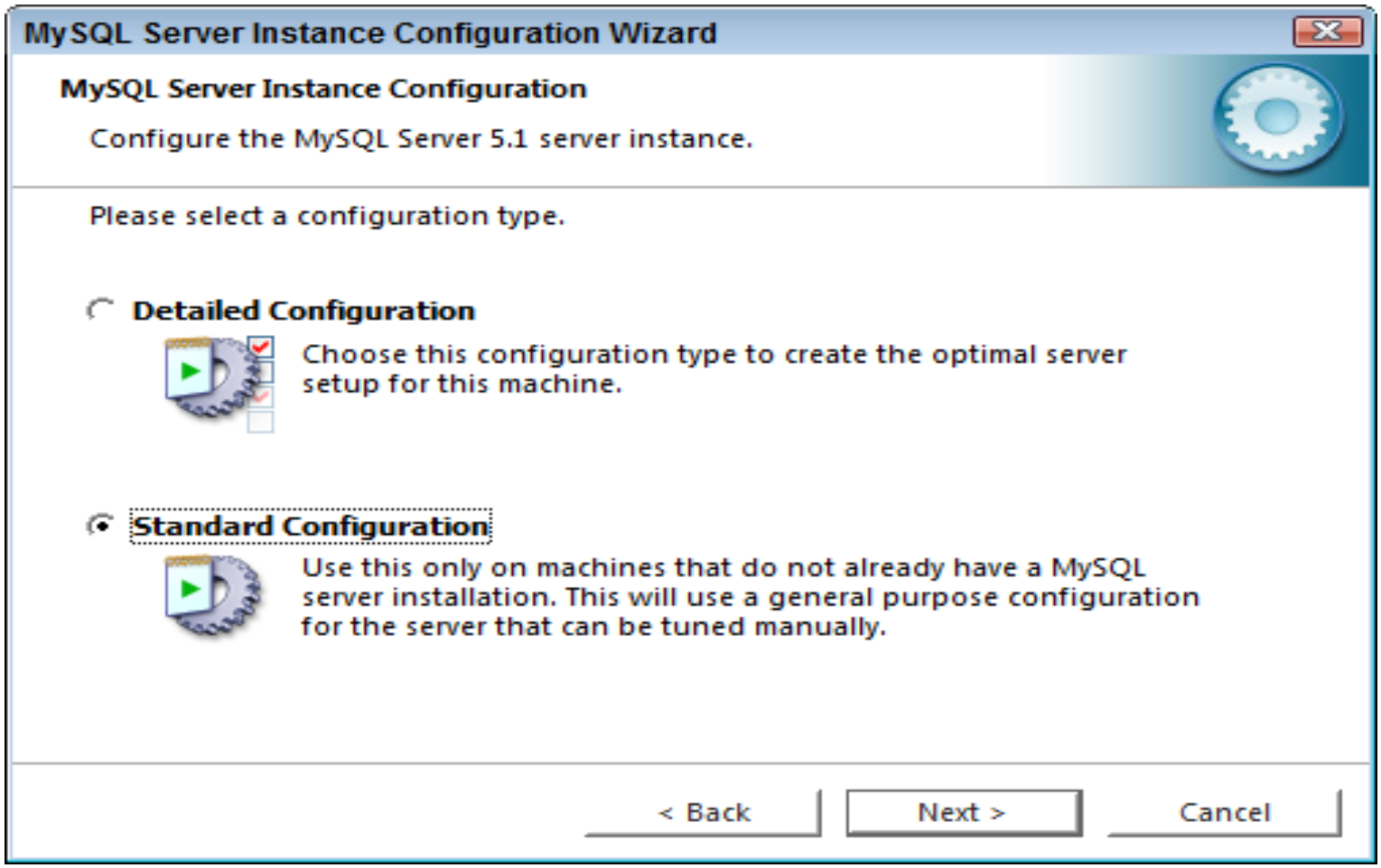
Click [Install] button



The installation can take some time depending on your computer's hardware configuration

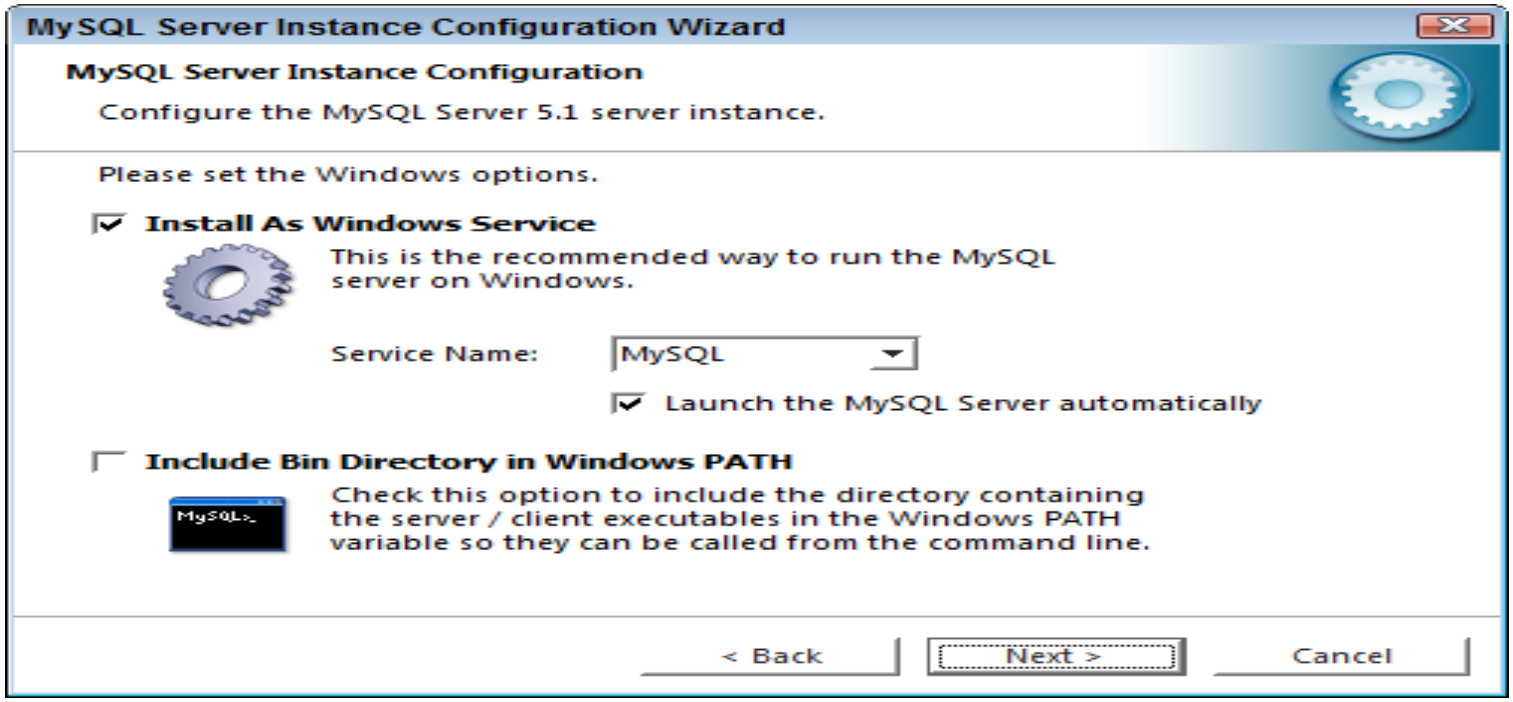
While configuration could be performed at later time, this tutorial does it as part of the installation process.
Click [Finish] button

Click [Next] button



Select the Standard Configuration option
Click [Next] button

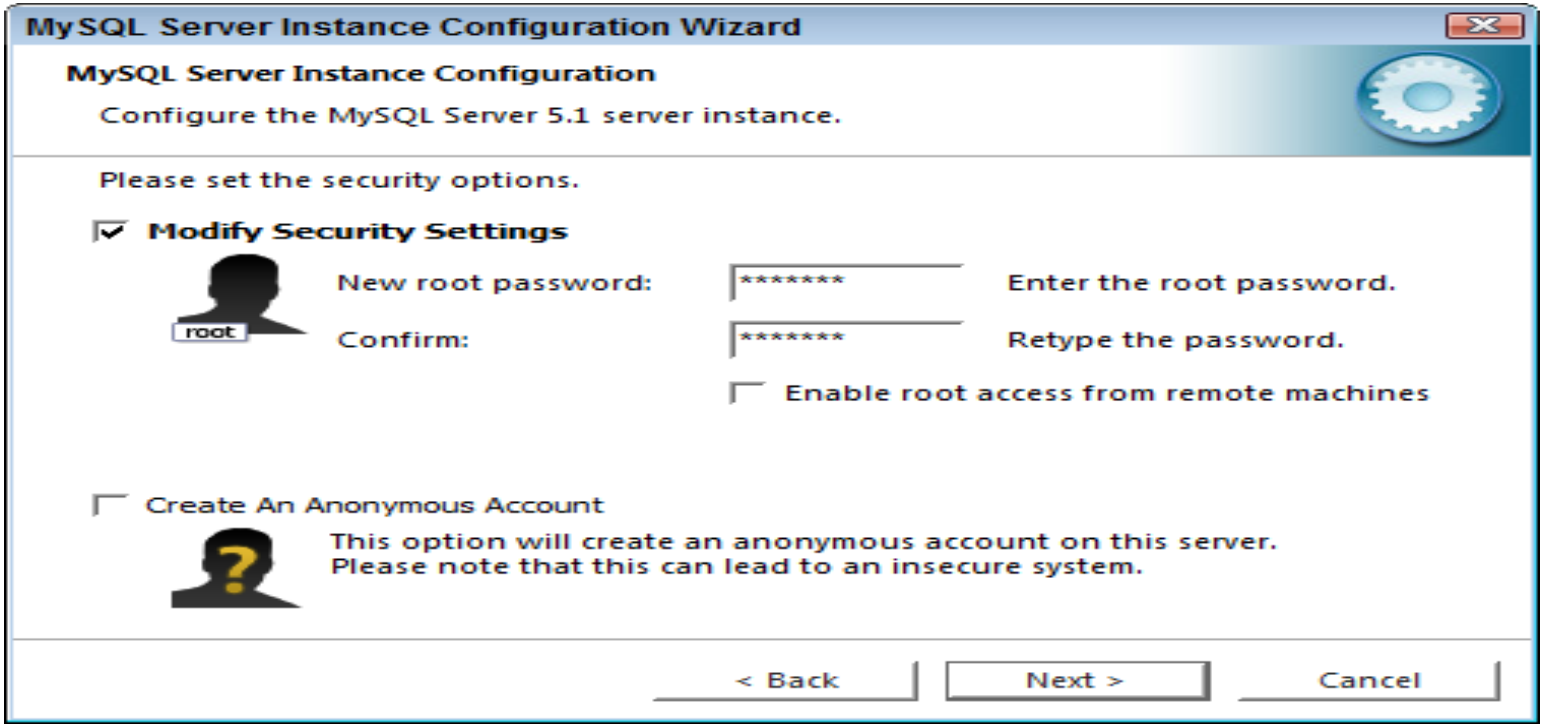
On Windows machines it is recommended to install server-type software as "Windows Service" as opposed to application. The advantages are that the MySQL server could start automatically on the machine startup, and will run in background thus utilizing fewer resources



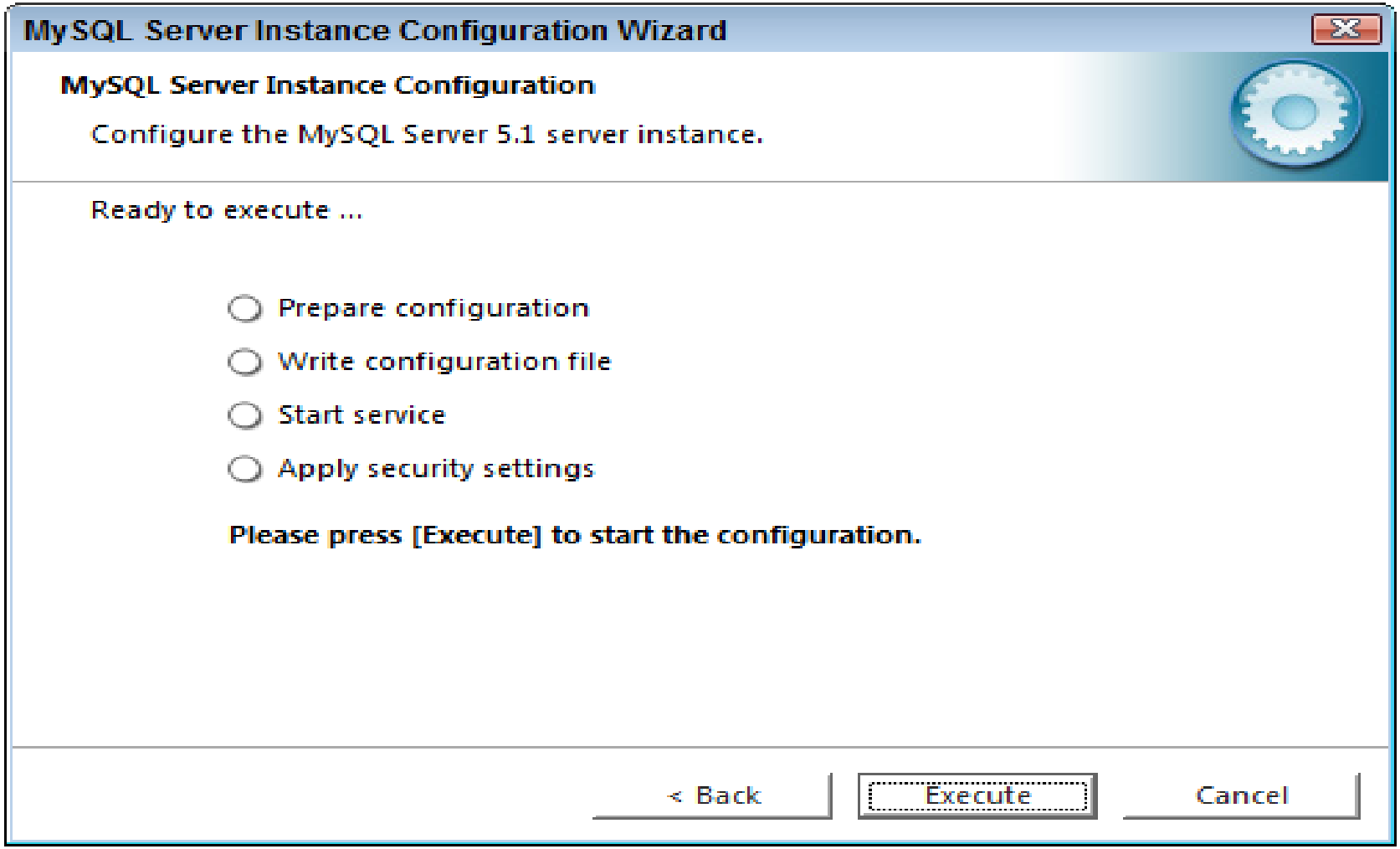
Select the Install As Windows Service option; check box to include MySQL \bin directory in Windows PATH environmental variable – this will facilitate using MySQL from command line.

Click [Next] button

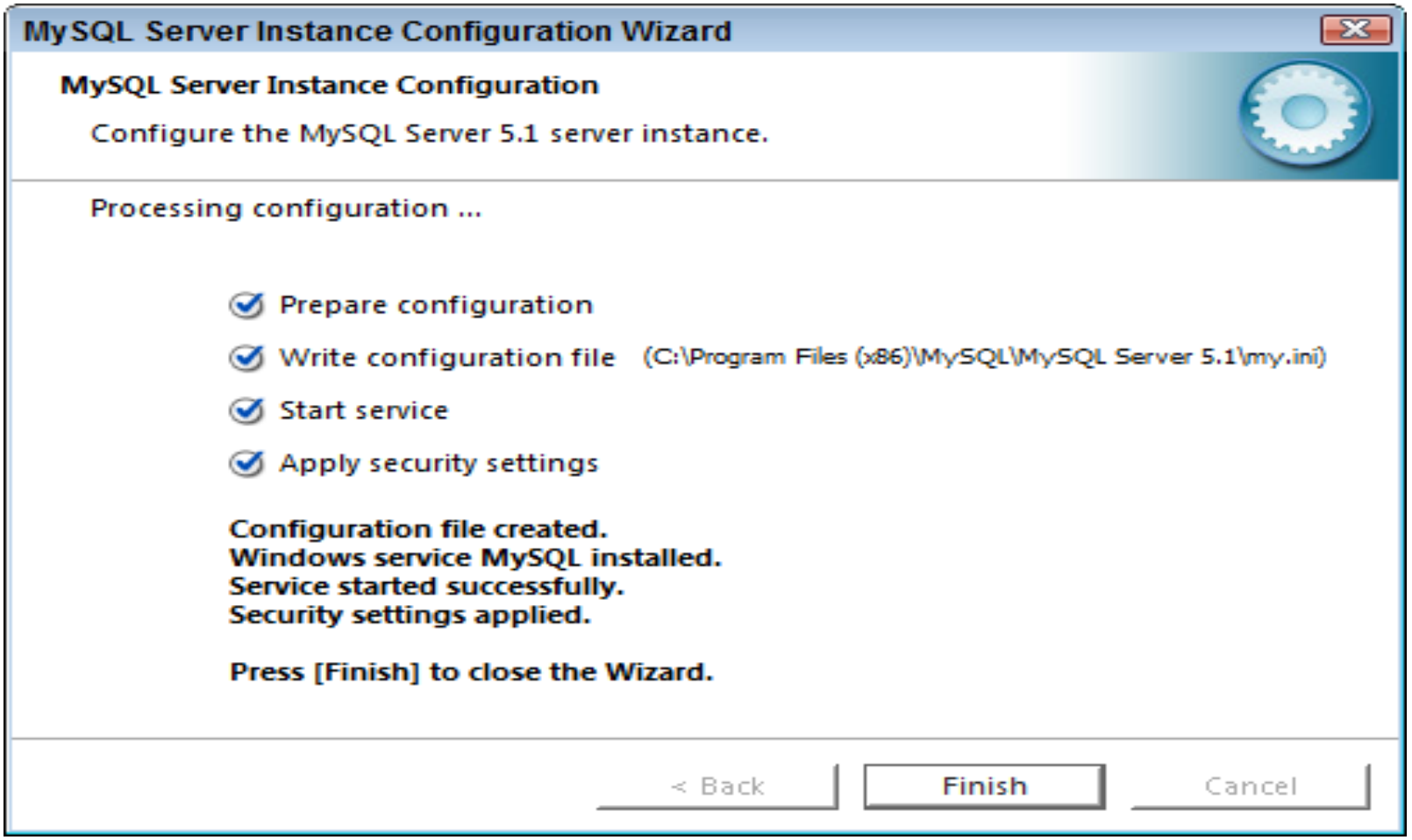
Password for administrative account is the key to the kingdom. Make it reasonably secure and safeguard it. Creating an Anonymous Account would allow anyone to connect to your database server without user ID and password; this could be potentially compromise security in your environment



Select the Modify Security Settings option and
select the root password
Click [Next] button



Click [Execute] button



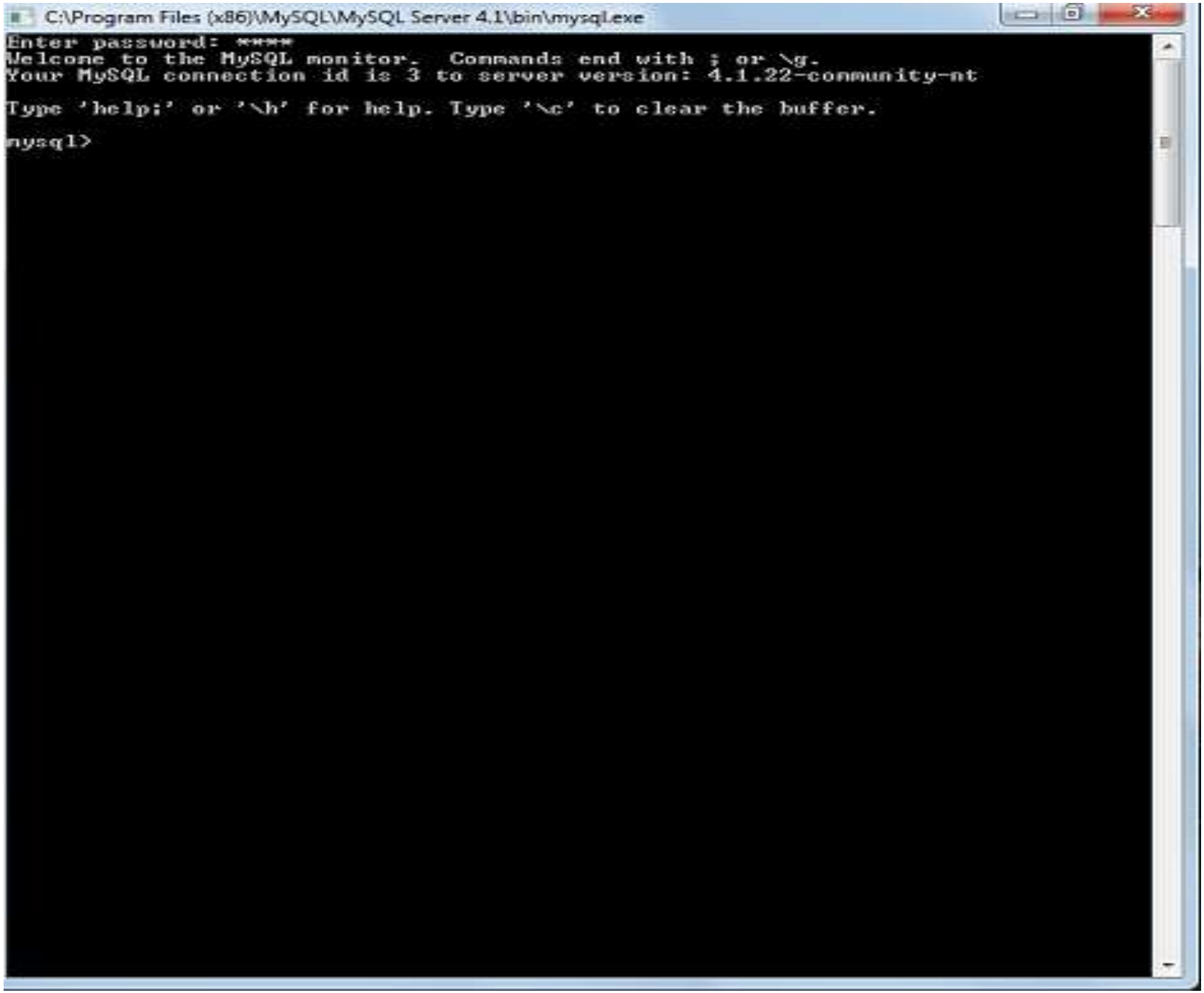
Click [Finish] button

Testing MySQL

Follow the steps to start MySQL

Start → All Programs → MySQL → MySQL Server → MySQL Command Line Client

MySQL will prompt a message to provide password (it requires the same password which was entered during the installation)


C:\Program Files (x86)\MySQL\MySQL Server 4.1\bin\mysql.exe
Enter password: ****
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 3 to server version: 4.1.22-community-nt
Type 'help;' or '\h' for help. Type '\c' to clear the buffer.
mysql>

MySQL and SQL

In order to access data within the MySQL database, all programs and user must use SQL(Structured Query Language). SQL is a Language that enables you to create and operate on relational databases.

Classification of SQL statements

- Data Definition Language (DDL) commands
- Data Manipulation Language (DML) commands
- Transaction Control Language (TCL) commands
- Session Control Commands
- System Control Commands

Data Definition Language (DDL) commands

It allows you to perform tasks related to data definition. You can perform tasks like:

- **Create, alter and drop schema objects**

  This section of DDL commands is used to create or define or change or delete objects such as a table, a view etc. e.g. CREATE TABLE, ALTER TABLE, DROP TABLE etc.

Data Manipulation Language (DML) commands

By data manipulation, we mean :

- the **retrieval** of the information stored in the database.
- the **insertion** of new information into the database
- the **deletion** of the information from the database.
- the **modification** of data stored in the database.

Data Manipulation Language (DML) commands

DML is a language that enables user to access or manipulate data as organized by the appropriate data model. E.g INSERT, UPDATE, DELETE etc.

Transaction Control Language (TCL) commands

These commands manages the changes made by DML commands. E.g.

BEGIN / START TRANSACTION : It marks the beginning of a transaction.

COMMIT: It ends the current transaction by saving database changes and start new transaction.

Transaction Control Language (TCL) commands

ROLLBACK: It ends the current transaction by discarding database changes and start new transaction.

SAVEPOINT: It defines breakpoints for the transaction to allow partial rollbacks.

SIMPLE QUERIES IN SQL

SQL

SQL was developed in 1970s in an IBM Laboratory. SQL sometimes also referred to as SEQUEL is 4^{th} generation non-procedural language.

MySQL SQL Elements

- Literals
- Datatypes
- Nulls
- Comments

Literals

It refer to a fixed data value. This fixed data value may be of character type or numeric literal. E.g. ‘Raj’ , ‘8’ , ‘305.8’ etc.

Data Types

Data Types are mean to identify the type of data and associated operations of handling it. Data Types in MySQL are divided into three categories:

- Numeric
- Date & Time
- String Type

Numeric Data Type

Common Numeric Data Types are:

- tinyint
- smallint
- mediumint
- int
- bigint

Numeric Data Type

Common Numeric Data Types are:

- float(m,d)
- double(m,d)
- decimal(m,d) or numeric(m,d)

Where ‘m’ is the total no. of digits and ‘d’ represent no. of decimals.

Date & Time Type

- DATE: stores date in ‘YYYY-MM-DD’ format.
- DATETIME: stores date & time in ‘YYYY-MM-DD HH-MM-SS’ format.
- TIMESTAMP: stores date & time in ‘YYYYMMDDHHMMSS’ format.
- TIME: stores time in ‘HH-MM-SS’ format.
- YEAR(M): stores a year in 2-digit or 4-digit format.

String/Text Types

- char(m)
- varchar(m)
- blob or text
- tinyblob or tinytext
- mediumblob or mediumtext
- longblob or longtext
- enum

Difference between char & varchar datatypes

char datatype

It specifies a fixed length character string. When a column is given a datatype as char(n), then MySQL ensures that all values stored in that column have this length i.e. n bytes. If a value is shorter than this length n then blanks are added, but the size of variable remains n bytes. Some space is always wasted as all the data elements do not use all the space reserved but processing is much simpler compare to varchar datatype.

varchar datatype

It specifies a variable length character string. When a column is given a datatype as varchar(n), then the maximum size a value in this column can have n bytes. If a value is shorter than this length n then no blanks are added, but the size of variable remains n bytes. There is no wastage of space but processing is much complex compare to varchar datatype.

Null Value

If a column in a row has no value, then column is said to be null or to contain a NULL.

Comments

A comment is a text that is not executed.

SQL provides following ways to write comment:

- ***/*** comment ******* **multiline**
- **--** comment single line
- # comment single line

SQL command syntax

- Keywords
- Commands or statements:- instructions given to sql database.
- clause: consist of command and arguments.

Making Simple Queries

- To understand these consider the following table in the database named “kvr”.

TABLE: EMP

Empid	empname	**empphone**	**empsal**	**deptno**
101	Ajay	234567	20000	25
102	Vijay	654378	15000	30
103	Ramesh	345678	10000	25
104	Ram	346279	15000	10

Accessing Database

To make queries upon the data in tables of a database, you need to open the database for use. SQL provide “use” command for this.

General Syntax:

use <databasename>;

To view tables in a database

SHOW TABLES;

The SELECT Command

It let you make query on the database. In Simplest form, SELECT statement is used as given below.

SELECT <columnname> [,<columnname>…] from <tablename>;

e.g SELECT empid, empname FROM emp;

SELECTING ALL COLUMNS

Genral syntax:

SELECT * FROM <tablename>;

e.g

SELECT * FROM emp;

REODRERING COLUMNS

Specify the name of column in same order as you want those to get display in SELECT command.

e.g.

e.g SELECT empname, empid FROM emp;

Eliminating Redundant Data

The DISTINCT keyword eliminates duplicate rows from the results of a SELECT statement.

e.g.

SELECT DISTINCT(deptno) FROM emp;

ALL keyword

ALL keyword retains the duplicate rows. It is just the same as when you specify neither DISTINCT nor ALL.

e.g.

SELECT ALL(deptno) FROM emp;

How to perform Simple Calculation

e.g.

SELECT 24*5;

DESC or DESCRIBE Command

Description: It is used to view the structure of table.

General Syntax:

DESC <tablename>;

or

DESCRIBE <tablename>;

Example:

DESC myemp;

Scalar Expressions with Selected Field

If you want to perform simple numeric computations on the data to put it in a form more appropriate to your needs, SQL allows you to place scalar expressions and constants among the selected fields.

e.g

SELECT empname, empsal+1000 FROM emp;

Using Column Aliases

It means that column you select in query can be given a different name.

e.g

SELECT empname, empsal*0.1 “Bonus” FROM emp;

Putting text in query Output

e.g

SELECT empname, ‘gets salary Rs.’ empsal
FROM emp;

RELATIONAL OPERATORS

To compare the two values SQL provides the following relational operator:

=

<=

>=

<

>

<>

Relational Operators

a	b	a<b	a<=b	a==b	a>b	a>=b	a!=b
0	0	false	true	true	false	true	false
0	1	true	true	false	false	false	true
1	0	false	false	false	true	true	true
1	1	false	true	true	false	true	false

LOGICAL OPERATORS

To combine the two conditions SQL provides the following logical operator:

OR (||)

AND (&&)

NOT (!)

Logical Operator

LOGICAL OR (||) operator: These return true only if any of the expression is true otherwise false.

LOGICAL AND (&&) operator: These return true only if both of the expression are true otherwise false.

LOGICAL NOT (!) operator: It negates the value of the expression.

SELECTING SPECIFIC ROWS

The WHERE clause in SELECT statement specifies the criteria for the selection of rows to be returned.

General Syntax:

SELECT <columnname> [,<columnname>…] from <tablename> where <condition>;

SELECTING SPECIFIC ROWS

e.g.

SELECT empname FROM emp WHERE empsal < 15000;

e.g.

SELECT empname FROM emp WHERE empid = 101;

Condition Based on a Range

The BETWEEN operator defines a range of values that the column values must fall in to make the condition true. The range includes both lower value and upper value.

e.g

SELECT empname FROM emp WHERE empsal BETWEEN 10000 and 15000;

Conditions based on a list

To specify a list of values, IN operator is used.

e.g

SELECT empname FROM emp WHERE deptno IN (’10’,’20’, ‘30’);

Conditions based on Pattern Matches

SQL provides “LIKE” operator for comparison on character strings using patterns. Patterns are described using two special **wildcard** characters.

- percent(%) : The % character matches any substring.
- underscore(_) : The _ matches any character.

Conditions based on Pattern Matches

e.g.

SELECT empid FROM emp WHERE empname LIKE “A%”;

e.g

SELECT empid FROM emp WHERE empname LIKE “_a”;

Searching for NULL

The NULL value in a column can be searched for in a table using “IS NULL” in the WHERE clause.

SELECT empname FROM emp WHERE deptno IS NULL;

Sorting Results

The ORDER BY clause allows sorting of query results by one or more columns.

e.g.

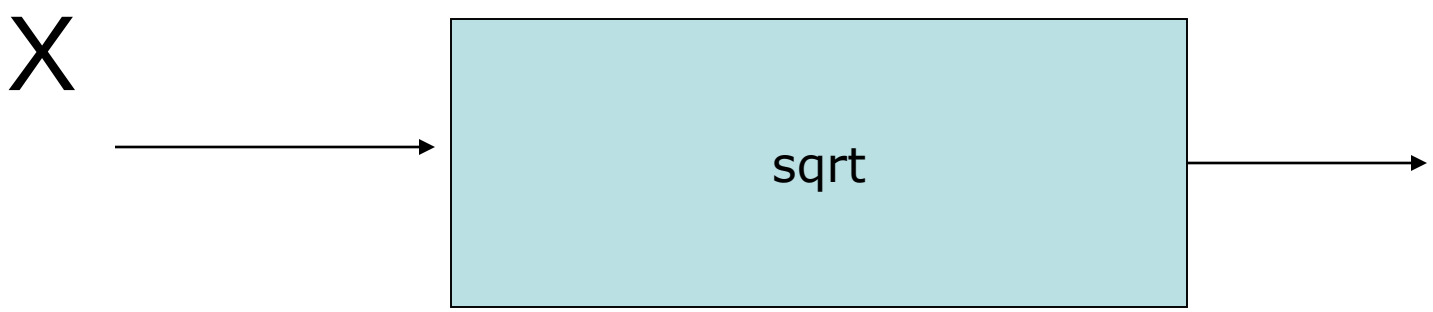
SELECT empname FROM emp ORDER BY empsal;

SQL FUNCTIONS
AND
TABLE JOINS

Function

- Function is a special type of predefined commands that performs some operation and return a single value.
- SQL Supports many and many Functions

Function sqrt



e.g

x=16 then y=4

Types of SQL Function

- (1)Single Row Function: works with data of single row.
- (2) Multiple Row or Group function: works with data of multiple rows.

Single Row function works with single data like sqrt function whereas multi row function work on more than one value like average function to find average of many numbers.

Multiple Row or Group function

TABLE: EMP

Empid	empname	**empphone**	**empsal**	**deptno**
101	Ajay	234567	20000	25
102	Vijay	654378	15000	30
103	Ramesh	345678	10000	25
104	Ram	346279	15000	10

AVG() function

- This function computes the average of given data.
- Suppose You want to find out the average of salary on EMP table then you can use AVG() function to do that

- e.g.
- SQL> SELECT AVG(empsal) “Average Salary” FROM EMP;

AVG() function

- Output:-
- Average Salary
- ----------
- 15000

COUNT() function

- This function computes the number of rows in a given column.
- Suppose You want to find out the number of employees in EMP table then you can use count() function to do that

- e.g.
- SQL> SELECT count(empid) “Total” FROM EMP;

COUNT() function

- Output:-
- Total
- ----------
- 5

MAX() function

- This function returns the maximum value for a given column.
- Suppose You want to find out the highest amount of salary paid from EMP table then you can use MAX() function to do that

MAX() function

- Output:-
- Maximum Salary
- ----------
- 20000

MIN() function

- This function returns the minimum value for a given column.
- Suppose You want to find out the lowest amount of salary paid from EMP table then you can use MIN() function to do that

- e.g.
- SQL> SELECT MIN(empsal) “Maximum salary” FROM EMP;

MIN() function

- Output:-
- Minimum Salary
- ----------
- 10000

SUM() function

- This function computes the sum of values of a given column.
- Suppose You want to find out the total salary paid to employees from EMP table then you can use SUM() function to do that

- e.g.
- SQL> SELECT SUM(empsal) “Total Salary” FROM EMP;

SUM() function

- Output:-
- Total Salary
- ----------
- 60000

GROUPING RESULTS

GROUP BY clause

The GROUP BY clause combines all those records that have identical values in a particular column. E.g. To calculate the number of employees in each department. You can use the command.

SELECT dept, count(*) FROM emp GROUP BY dept;

GROUP BY clause

- Output:-

deptno	count(*)
25	2
30	1
10	1

HAVING Clause

The having clause places conditions on groups.

e.g. To calculate the number of employees in department no. 25. You can use the command.

SELECT dept, count(*) FROM emp GROUP BY dept HAVING dept = 25;

HAVING Clause

- Output:-

deptno	count(*)
25	2

Cartesian product

The **Cartesian product**, also referred to as a cross-join, returns all the rows in all the tables listed in the query. Each row in the first table is paired with all the rows in the second table.

The Cartesian Product of Two Relations (cont.)

- Example:

Enrolled

student_id	*course_name*	*credit_status*
12345678	CS 105	ugrad
25252525	CS 111	ugrad
45678900	CS 460	grad
33566891	CS 105	non-credit
45678900	CS 510	grad

MajorsIn

student_id	*dept_name*
12345678	comp sci
45678900	mathematics
25252525	comp sci
45678900	english
66666666	the occult

Enrolled x MajorsIn

Enrolled. student_id	*course_name*	*credit_status*	*MajorsIn. student_id*	*dept_name*
12345678	CS 105	ugrad	12345678	comp sci
12345678	CS 105	ugrad	45678900	mathematics
12345678	CS 105	ugrad	25252525	comp sci
12345678	CS 105	ugrad	45678900	english
12345678	CS 105	ugrad	66666666	the occult
25252525	CS 111	ugrad	12345678	comp sci
25252525	CS 111	ugrad	45678900	mathematics
...	...	...	...	...

Joins

A join is a query that combines two or more tables.

For example:

select * from emp, dept;

will give all the possible combination formed of all the rows of both the tables.

Equi Join

The join in which columns are compared for equality is called equi-join.

Non-equi Join

It is a query that specifies some relationship other than equality between columns of tables.

Natural Join

The Join in which only one of the identical column exist is called natural join.

CROSS JOIN

It is a very basic type of join that simply matches each row from one table to every row from another table.

DBMS Concepts

Database

A database may be defined as a collection of interrelated data stored together to serve multiple applications.

DBMS

A DBMS(Data Base Management System) refers to a software that is responsible for storing, maintaining and utilizing databases.

Database System

A database along with a DBMS is referred to as a database system.

Advantages of Database System

- Databases reduces the data redundancy to a large extent.

Data Redundancy

Duplication of data is known as data redundancy.

Office
Shiv
Panipat
Haryana
Office ← -------Hostel
Shiv
Panipat
Haryana

Hostel
Shiv
Rohtak
Haryana

Advantages of Database System

- Databases can control data inconsistency to a large extent.
- Databases facilitate sharing of data.
- Databases enforce standards.
- Databases can ensure data security.

Data Security

It refers to the protection of data against accidental or intentional disclosure to unauthorized persons, or unauthorized modification or destruction.

Privacy of Data

It refers to the rights of individuals and organizations to determine themselves when, how, and to what extent information about them is to be transmitted to other.

Advantages of Database System

- Integrity can be maintained through databases.

Data Model

It refers to a set of concepts to describe the structure of a database and certain constraints that the database should obey. Different data models are:

- Relational Data Model
- Hierarchical Data Model
- Network Data Model
- Object Oriented Data Model

Relational Data Model

It was propounded by E.F. Codd of the IBM.

In relational data model, the data is organized into tables(i.e. rows and columns). These tables are called relations.

Relational Data Model Terminology

View

It is a table that does not exist in its own right but is instead derived from one or more underlying base tables.

Degree

It is number of attributes in a table or relation.

Cardinality

It is number of tuples/records in a table or relation.

Primary Key

It is a set of one or more attributes that can uniquely identify tuples within the relation.

Candidate Key

All attribute combinations inside a relation that can serve as primary key are candidate keys.

Alternate Key

A candidate key that is not the primary key is called an alternate key.

Foreign Key

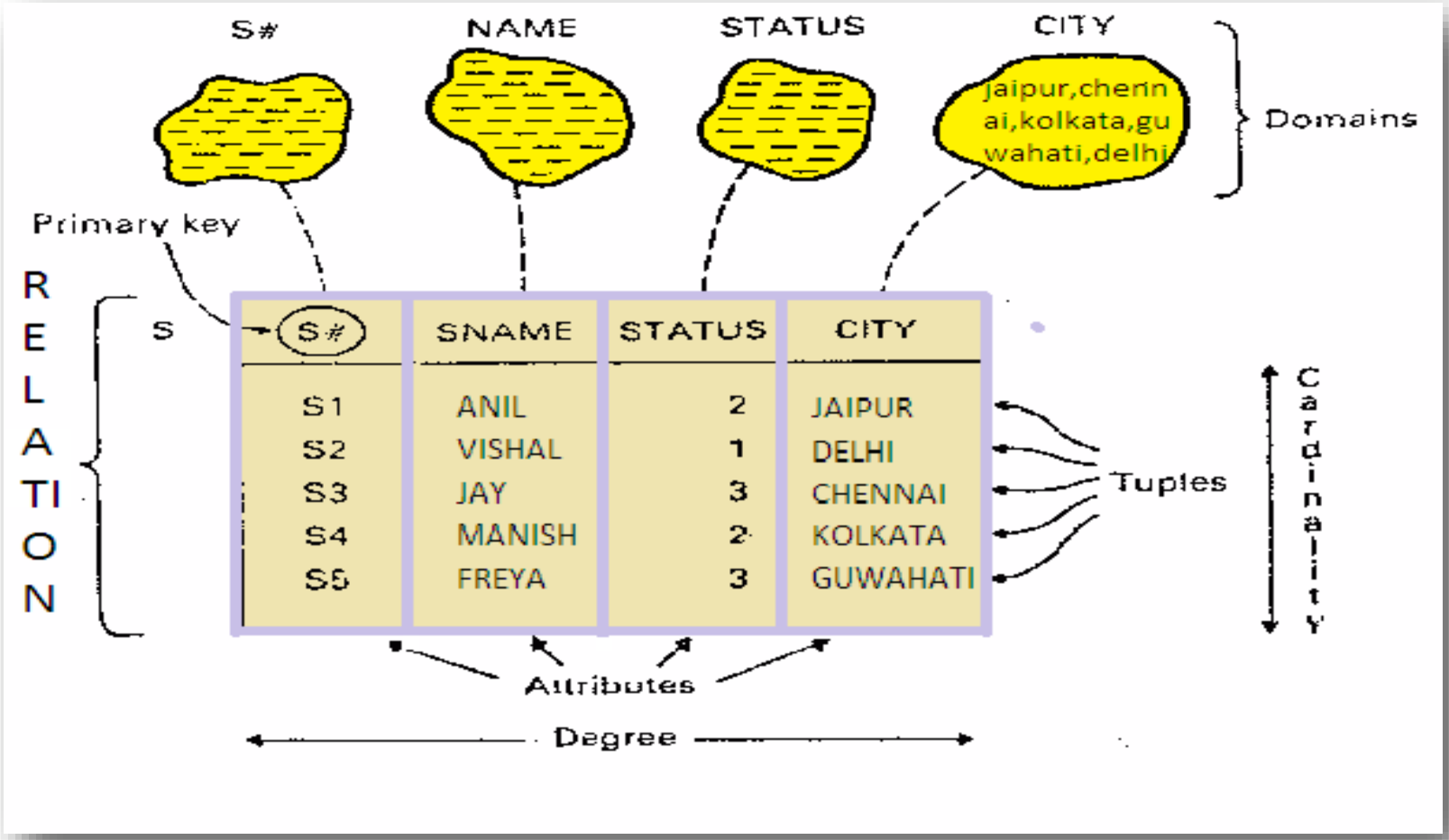
A non-key attribute, whose values are derived from the primary key of some other table, is known as Foreign-Key in the current table.

Referential Integrity

It is a system of rules that DBMS uses to ensure that relationships between records in related tables are valid, and that users don’t accidentally delete or change related data.

DATABASE CONCEPTS

RELATIONAL DATABASE TERMS



DATABASE CONCEPTS

KEYS IN A DATABASE

Super Key

Primary Key

Candidate Key

Unique Key

Alternate Key

Employee Table

Employee_Id	Employee_Name	Address	License_Number	Passport_Number
100	FREYA	121,AJMER	AJ38839929	PS939303002
101	MOHAK	4/2B JAIPUR	JP93930030	P3392929393
102	RAJESH	3 UDAIPUR	UP3839392	P3939392020

Composite Key

Foreign Key

Salary Table

Employe_Id	Salary_Month_Year	Amount
100	2018 -Jan	10000
101	2018 -Jan	11000
102	2018 -Jan	15000
100	2018 -Feb	10000
101	2018 -Feb	11000
102	2018 -Feb	15000